

Lab 3 Individual Proof

Task 2:

Terminal Results from Step 8:

```
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$  
colcon build --symlink-install  
Starting >>> webcam  
--- stderr: webcam  
/usr/lib/python3/dist-packages/setuptools/command/easy_install.py:158: EasyInstallDeprecationWarn  
ing: easy_install command is deprecated. Use build and pip and other standards-based tools.  
  warnings.warn(  
---  
Finished <<< webcam [4.22s]  
  
Summary: 1 package finished [6.27s]  
1 package had stderr output: webcam
```

Task 3:

Results from Running the Commands of Step 22:

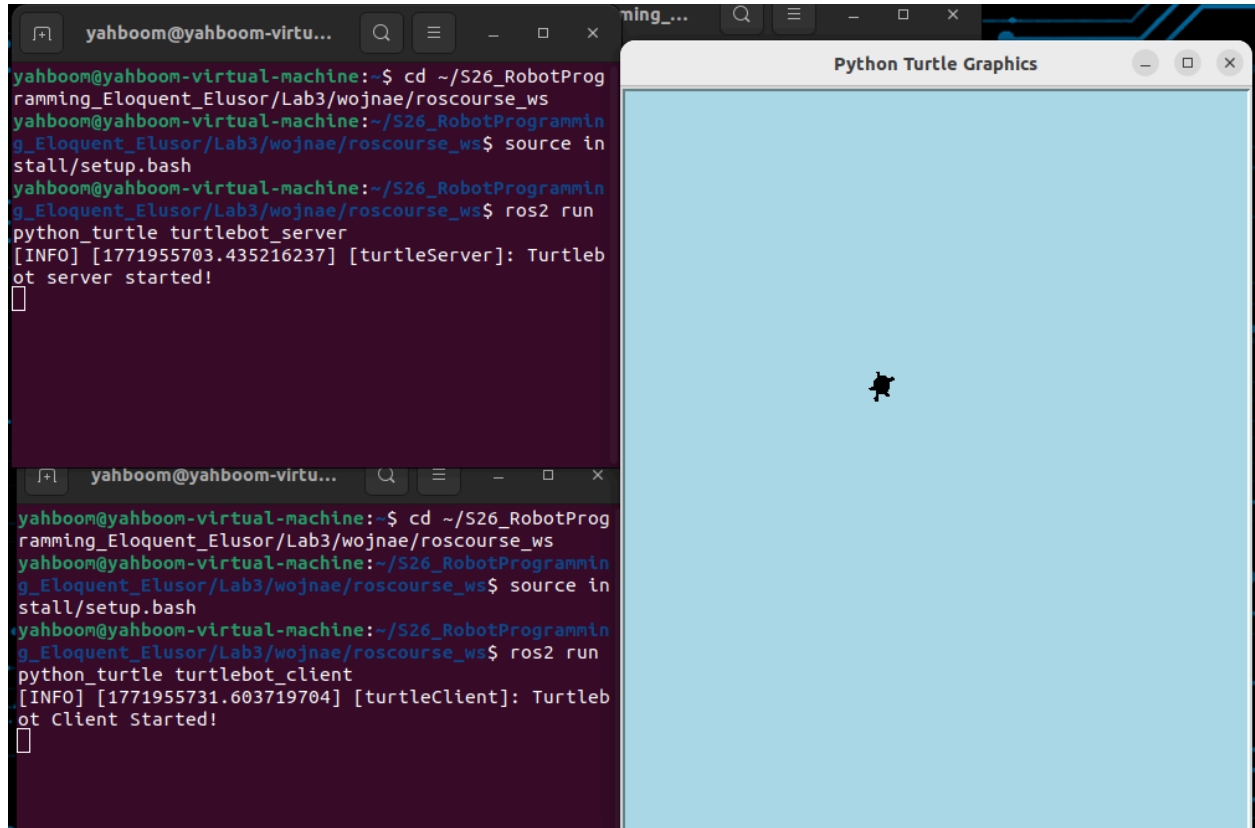
```
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/r  
oscourse_ws$ ros2 interface show turtle_interfaces/msg/TurtleMsg  
string name  
geometry_msgs/Pose turtle_pose  
  Point position  
    float64 x  
    float64 y  
    float64 z  
  Quaternion orientation  
    float64 x 0  
    float64 y 0  
    float64 z 0  
    float64 w 1  
string color
```

```
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/r
oscourse_ws$ ros2 interface show turtle_interfaces/srv/SetPose
geometry_msgs/PoseStamped turtle_pose
  std_msgs/Header header
    builtin_interfaces/Time stamp
      int32 sec
      uint32 nanosec
    string frame_id
  Pose pose
    Point position
      float64 x
      float64 y
      float64 z
    Quaternion orientation
      float64 x 0
      float64 y 0
      float64 z 0
      float64 w 1
---
int8 ret
```

```
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/r
oscourse_ws$ ros2 interface show turtle_interfaces/srv/SetColor
string color
---
int8 ret
```

Task 4:

Results from Step 7:

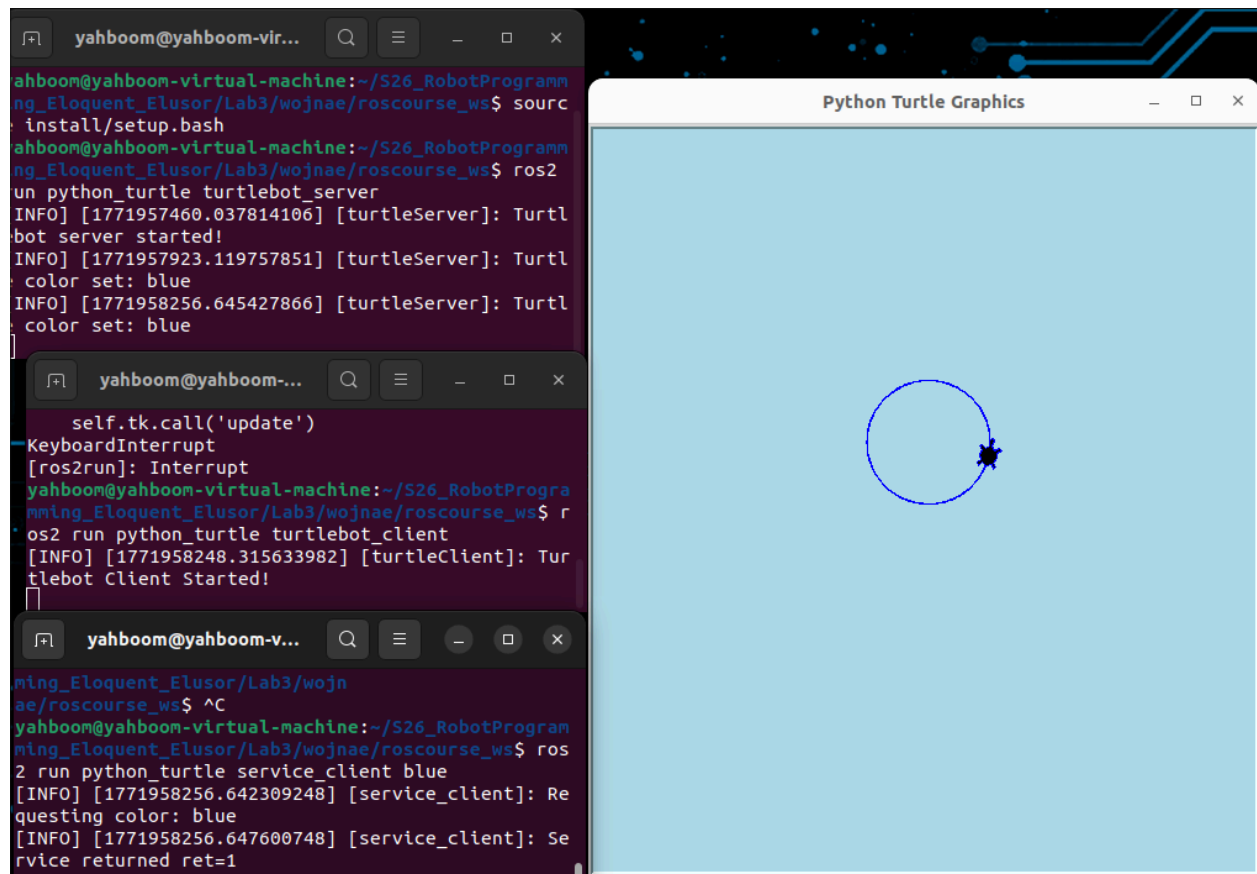


The screenshot shows two windows. The left window is a terminal with a dark purple background, displaying the execution of ROS2 commands to start a turtlebot server and client. The right window is titled 'Python Turtle Graphics' and shows a light blue canvas with a small black turtle icon in the center.

```
yahboom@yahboom-virtual-machine:~$ cd ~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ ros2 run python_turtle turtlebot_server
[INFO] [1771955703.435216237] [turtleServer]: Turtlebot server started!
□

yahboom@yahboom-virtual-machine:~$ cd ~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ ros2 run python_turtle turtlebot_client
[INFO] [1771955731.603719704] [turtleClient]: Turtlebot Client Started!
□
```

Results from Step 16:



The image shows a terminal window and a Python Turtle Graphics window. The terminal window displays the following commands and output:

```
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ source install/setup.bash
yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ ros2 run python_turtle turtlebot_server
INFO [1771957460.037814106] [turtleServer]: Turtlebot server started!
INFO [1771957923.119757851] [turtleServer]: Turtlebot color set: blue
INFO [1771958256.645427866] [turtleServer]: Turtlebot color set: blue

yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ ros2 run python_turtle turtlebot_client
[INFO] [1771958248.315633982] [turtleClient]: Turtlebot Client Started!

yahboom@yahboom-virtual-machine:~/S26_RobotProgramming_Eloquent_Elusor/Lab3/wojnae/roscourse_ws$ ros2 run python_turtle service_client blue
[INFO] [1771958256.642309248] [service_client]: Requesting color: blue
[INFO] [1771958256.647600748] [service_client]: Service returned ret=1
```

The Python Turtle Graphics window shows a blue circle drawn on a light blue background. A small black turtle icon is positioned at the end of the circle, indicating it has completed the drawing.